

Aco numerical

Key Concepts in Ant Colony Algorithm

1. Ants as Agents:

- Artificial "ants" are used to simulate the behavior of real ants.
- Each ant represents a possible solution to the problem.

2. Graph Representation:

- The problem is represented as a graph $G = (N, E)$, where:
 - N : Set of nodes (e.g., cities in TSP).
 - E : Set of edges (connections between nodes with associated costs or distances).

3. Pheromone:

- A numeric value is associated with each edge of the graph, representing the desirability of that edge.

4. Heuristic Information:

- A heuristic η_{ij} is often used to guide the ants (e.g., the inverse of the distance between two nodes).

5. Probability of Choosing an Edge:

- An ant chooses the next node to visit based on a probabilistic rule:

$$P_{ij} = \frac{\tau_{ij}^{\alpha} \cdot \eta_{ij}^{\beta}}{\sum_{k \in \text{allowed}} \tau_{ik}^{\alpha} \cdot \eta_{ik}^{\beta}}$$

- τ_{ij} : Pheromone level on edge (i, j) .
- η_{ij} : Heuristic information.
- α : Importance of pheromone.
- β : Importance of heuristic.
- **allowed**: Set of nodes that can be visited next.

6. Pheromone Update:

- After all ants complete their solutions, the pheromone levels are updated:

- Evaporation:

$$\tau_{ij} = (1 - \rho) \cdot \tau_{ij}$$

where ρ (evaporation rate) controls the decay of pheromone.

- Reinforcement:

$$\tau_{ij} = \tau_{ij} + \sum_{\text{ants}} \Delta\tau_{ij}$$

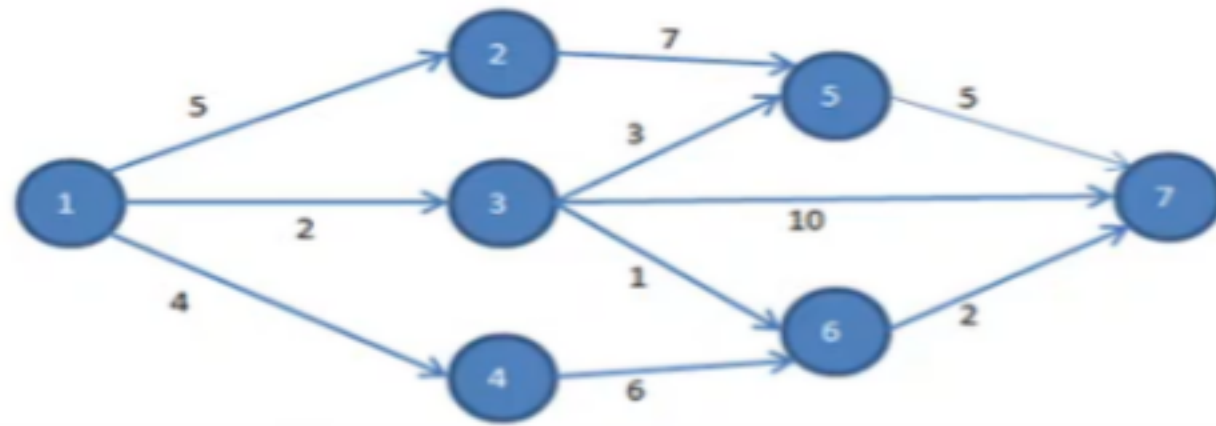
where $\Delta\tau_{ij}$ represents the pheromone deposited by ants that used edge (i, j) :

$$\Delta\tau_{ij} = \begin{cases} Q/L_k & \text{if edge } (i, j) \text{ is used by ant } k \\ 0 & \text{otherwise} \end{cases}$$

Q : Constant; L_k : Total cost of the solution found by ant k .

7. Iterative Process:

- The process of ants building solutions and updating pheromones is repeated over multiple iterations, gradually converging to high-quality solutions.



Path	Phrm	dist	1/dist	phrm*1/dist
1>2	1	5	0.2	0.2
1>3	1	2	0.5	0.5
1>4	1	4	0.25	0.25
2>5	1	7	0.14286	0.142857143
3>5	1	3	0.33333	0.333333333
3>6	1	1	1	1
3>7	1	10	0.1	0.1
4>6	1	6	0.16667	0.166666667
5>7	1	5	0.2	0.2
6>7	1	2	0.5	0.5

Node	Path	Prob	cum. Prob	R1#	R2#	R3#
1	1>2	0.21053	0.21053	0.223888	0.673577	0.991243
	1>3	0.52632	0.73684	Ant1 Path	Ant2 Path	Ant3 Path
	1>4	0.26316	1.00000	1>3	1>3	1>4

Node	Path	Prob	cum. Prob	R1#	R2#
3	3>5	0.23256	0.23256	0.169776	0.987414
	3>6	0.69767	0.93023	Ant1 Path	Ant2 Path
	3>7	0.06977	1.00000	3>5	3>7

Node	Path	Prob	cum. Prob	R3#
4	4>6	1.00000	1.00000	0.170916
				Ant3 Path
				4>6

Node	Path	Prob	cum. Prob	R1#
5	5>7	1.00000	1.00000	0.483519
				Ant1 Path
				5>7

Node	Path	Prob	cum. Prob	R3#
6	6>7	1.00000	1.00000	0.388501
				Ant3 Path
				6>7

$$P_{i,j} = \frac{(\tau_{i,j})^\alpha (\eta_{i,j})^\beta}{\sum \left((\tau_{i,j})^\alpha (\eta_{i,j})^\beta \right)}$$

$P_{i,j} \rightarrow$ Probability of choosing path i, j

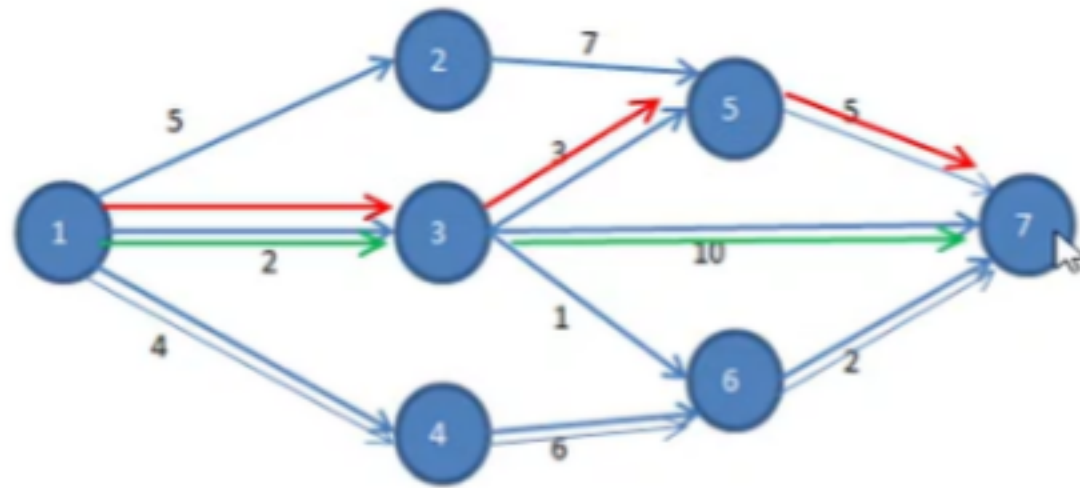
$\tau_{i,j} \rightarrow$ Pheromone on path i, j

$$\eta_{i,j} \rightarrow \frac{1}{L_{i,j}}$$

Pherm $\rightarrow$ Pheromone

Ri# $\rightarrow i^{th}$ random number

	Path	L_k	$1/L_k$
Ant1 path	1-3-5-7	10	0.1
Ant2 Path	1-3-7	12	0.08333333
Ant3 Path	1-4-6-7	12	0.08333333



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1>2	1	5	0.2	0.2
1>3	1	2	0.5	0.5
1>4	1	4	0.25	0.25
2>5	1	7	0.14286	0.142857143
3>5	1	3	0.33333	0.333333333
3>6	1	1	1	1
3>7	1	10	0.1	0.1
4>6	1	6	0.16667	0.166666667
5>7	1	5	0.2	0.2
6>7	1	2	0.5	0.5

Path		L_k	$1/L_k$
Ant1 path	1-3-5-7	10	0.1
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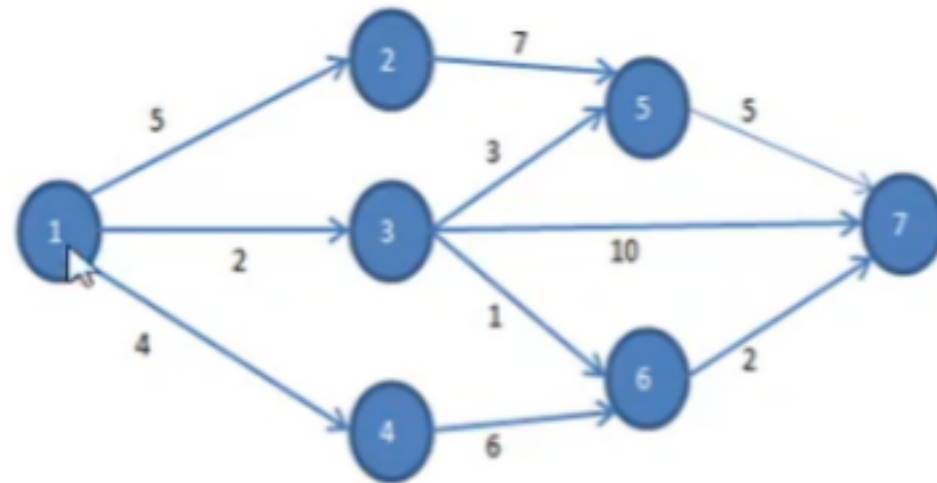
Path	Phrm balance	Phrm added	Tot. Phrm	dist	1/dist	phrm*1/dist
1>2	0.5	0	0.5	5	0.2	0.1
1>3	0.5	0.1833	0.6833	2	0.5	0.34165
1>4	0.5	0.0833	0.5833	4	0.25	0.145825
2>5	0.5	0	0.5	7	0.142857	0.071429
3>5	0.5	0.1	0.6	3	0.333333	0.2
3>6	0.5	0	0.5	1	1	0.5
3>7	0.5	0.0833	0.5833	10	0.1	0.05833
4>6	0.5	0.0833	0.5833	6	0.166667	0.097217
5>7	0.5	0.1	0.6	5	0.2	0.12
6>7	0.5	0.0833	0.5833	2	0.5	0.29165

$(1 - \rho) * \text{Pheromone on path}$

$\rho = \text{Evaporation Rate}$

$k \rightarrow \text{Ant number}$

Sum of $\frac{1}{L_k}$ for all ants k on arc i, j



Node	Path	Prob	cum. Prob	R1#	R2#	R3#
1	1>2	0.17022	0.17022	0.868596	0.635028	0.541652
	1>3	0.58156	0.75178	Ant1 Path	Ant2 Path	Ant2 Path
	1>4	0.24822	1.00000	1>4	1>3	1>3

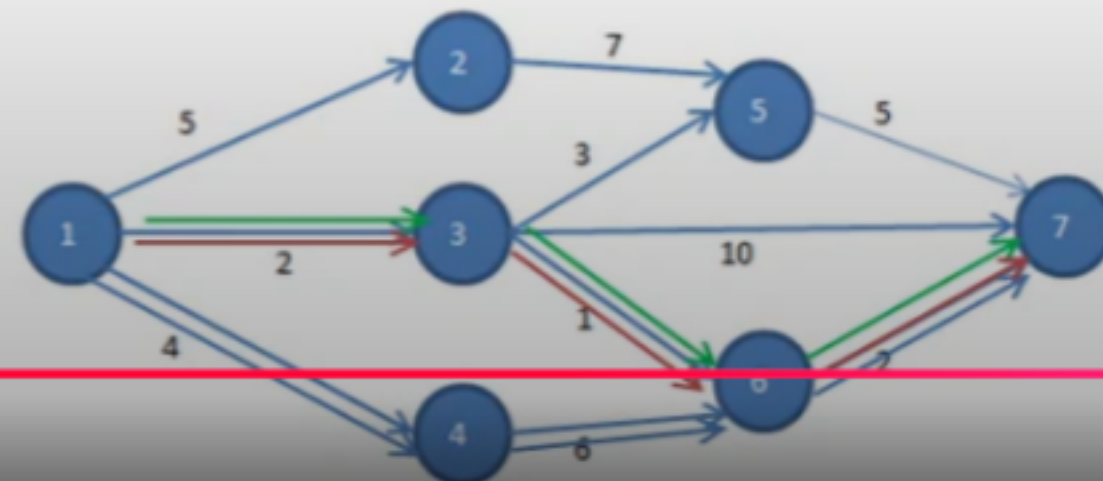
Node	Path	Prob	cum. Prob	R2#	R3#
3	3>5	0.26374	0.26374	0.553541	0.787063
	3>6	0.65934	0.92308	Ant1 Path	Ant2 Path
	3>7	0.07692	1.00000	3>6	3>6

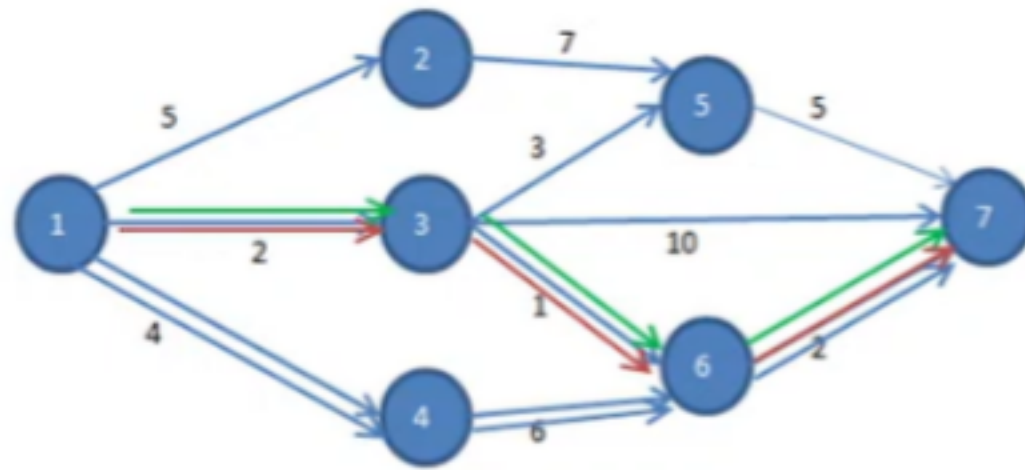
Node	Path	Prob	cum. Prob	R3#
4	4>6	1.00000	1.00000	Ant3 Path
				4>6

Node	Path	Ant3 Path
6	6>7	6>7

Path	Phrm balance	Phrm added	Tot. Phrm	dist	1/dist	phrm*1/dist
1>2	0.5	0	0.5	5	0.2	0.1
1>3	0.5	0.1833	0.6833	2	0.5	0.34165
1>4	0.5	0.0833	0.5833	4	0.25	0.145825
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3>6	0.5	0	0.5	1	1	0.5
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6>7	0.5	0.0833	0.5833	2	0.5	0.29165

	Path	L_k	$1/L_k$
Ant1 path	1-4-6-7	12	0.08333333
Ant2 Path	1-3-6-7	5	0.2
Ant3 Path	1-3-6-7	5	0.2





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1>2	0.5	0	0.5	5	0.2	0.1
1>3	0.5	0.1833	0.6833	2	0.5	0.34165
1>4	0.5	0.0833	0.5833	4	0.25	0.145825
2>5	0.5	0	0.5	7	0.142857	0.071429
3>5	0.5	0.1	0.6	3	0.333333	0.2
3>6	0.5	0	0.5	1	1	0.5
3>7	0.5	0.0833	0.5833	10	0.1	0.05833
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6>7	0.5	0.0833	0.5833	2	0.5	0.29165

Path		L_k	$1/L_k$
Ant1 path	1-4-6-7	12	0.08333333
Ant2 Path	1-3-6-7	5	0.2
Ant3 Path	1-3-6-7	5	0.2

Process continues. All ants finally follow path 1-3-6-7. That is the shortest path

Path	Phrm Balance	Phrm add	Tot. Phrm	dist	1/dist	phrm*1/dist
1>2	0.25	0	0.25	5	0.2	0.05
1>3	0.34165	0.4	0.74165	2	0.5	0.370825
1>4	0.29165	0.0833	0.37495	4	0.25	0.093738
2>5	0.25	0	0.25	7	0.142857	0.035714
3>5	0.3	0	0.3	3	0.333333	0.1
3>6	0.25	0.4	0.65	1	1	0.65
3>7	0.29165	0	0.29165	10	0.1	0.029165
4>6	0.29165	0.0833	0.37495	6	0.166667	0.062492
5>7	0.3	0	0.3	5	0.2	0.06
6>7	0.29165	0.4833	0.77495	2	0.5	0.387475

Consider a Traveling Salesman Problem (TSP) where you need to visit 5 cities, starting and ending at city A. The distance matrix between the cities is as follows:

City	A	B	C	D	E
A	0	10	15	20	25
B	10	0	35	25	30
C	15	35	0	30	5
D	20	25	30	0	15
E	25	30	5	15	0

The goal is to find the shortest path that visits each city exactly once and returns to the starting city. Solve TSP using Ant Colony Optimization (ACO). Given the evaporation factor $\rho = 0.02$ and initial pheromone at all edges $\tau_{ij} = 100$.
